NatiFine SA

Digital Traffic Fines System

Project Name: NatiFine SA – Digital Traffic Fines System.

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3. Analysis Phase

3.1 Introduction

The Planning Phase (Phase 2) established that traffic fine administration in South Africa is fragmented, address-based and largely manual, and it confirmed that a centralised, ID-linked prototype called NatiFine SA is necessary, technically achievable, operationally beneficial and economically viable. Phase 3 builds on those findings. Whereas Phase 2 answered the question “should we build this and how will we organise the work?”. The Analysis Phase answers, “what exactly must the system do, and what must it store?”.

The purpose of this phase is to study the current way in which fines are issued, communicated, paid and disputed and to identify the data the proposed system must hold and the rules that keep that data correct, to expose the weaknesses of the current arrangement, and to translate these findings into clear functional and non-functional requirements and a data model. The output of this phase is the input to the Design phase, so every requirement is written to be testable.

This chapter is organized as follows:

- 3.2 Information gathering methodology: how facts about the current system were collected.
- 3.3 Analysis of the existing system: how fines are handled today.
- 3.4 Data analysis: the data to be stored, its integrity rules and constraints.
- 3.5 Weaknesses of the current system.
- 3.6 Analysis of the proposed system: functional requirements.
- 3.7 Non-Functional requirements.
- 3.8 Data modelling for the proposed system.

Scope reminder. As agreed in Phase 2, NatiFine SA is an academic prototype built with Python, HTML/CSS, JavaScript and SQLite. Payments, SMS and WhatsApp delivery and government data are simulated whereas all fines, motorists and payment details are used for demonstration and are fictitious.

3.2 Information Gathering Methodology

The methodology for the project NatiFine SA is based on preliminary research, analysis of available information in public domain, as well as evaluation of needs identified in Phases 1 and 2. There were no actual interviews with motorists, traffic officers or municipal officials carried out. No interview outcomes were presented or indicated that feedback was gained from external representatives in the analysis. Instead, project documents and information available in public domain were utilized for problem understanding and project requirements identification.

The methodology proved to be quite relevant for the academic purposes of this project, as the educators had to understand the concept of traffic fines, point out major weaknesses and figure out basic functions of the prototype.

3.2.1 Observation

The team assessed the problem statement, project proposal and planning documentation which were prepared in the First and Second phases of the project. This documentation stated that fragmented activities are accompanied by problems, and a solution was proposed in the form of the required digital platform where drivers will be able to see their fines, perform payment simulations, file complaints and get notifications.

Publicly available information on the traffic violations in South Africa was used in order to get an idea of the basic stages of dealing with the infringement notifications. The guidance on the Administrative Adjudication of Road Traffic Offences published by the South African Government states such processes as making payments and declaring the incident (Government, 2010). Such information was useful while analyzing the processes that will be demonstrated in the prototype.

The disadvantage of the given method is that public materials do not give insight into local internal procedures, nor do they guarantee that all government bodies have the same procedures. As a result, the derived knowledge is going to be used in the process of establishing the requirements of the prototype but not as an argument for claiming that every traffic authority operates identically.

3.2.2 Participatory Method

The project group made an analysis of issues and targets as described in Phase 1 documentation and Phase 2 design papers. These issues included the problem of places where fines could be displayed, a need for the clear payment and dispute system, the necessity for reminders and accessible information about fines.

Then all the requirements were transformed into system functions, such as the registration of motorists and login, fine display, demonstration of payment operations, submission of disputes, sending notifications and reminders, and interaction with an admin. The limitations of the project were kept in mind as well – the use of fictional data and lack of real government databases, banking and SMS/WhatsApp gates.

This way, the team made sure that their analysis was consistent with the official project outline and the Agile methodology in question. However, at this stage, as formal interviews were not conducted, and external user testing was not performed yet, the project group cannot state that the requirements were validated with the help of representative groups.

3.2.3 Interviews

The primary disadvantage was the lack of formal interviews and direct confirmations from the municipal traffic authorities. As a result, the team was not able to verify independently all user experiences or internal administrative procedures. The results are based on the materials gained from the project and available public information and hence should be taken in this context.

It would be possible to acquire further feedback during a later stage of the Agile development process by approaching the supervisor and, if feasible, prospective users regarding the usability of the prototypes. Any feedback obtained will be recorded accordingly and used to adjust the requirements.

Summary of Findings

The preliminary research has shown that there is a need in the market for a solution to integrate key functions of fine management to one interface. The proposed technology will allow drivers to see the fines associated with their accounts, simulate payment processes, make claims and receive notifications. An officer or an administrator will be tasked with entering fines and managing disputes.

The findings are based on initial project research and document review and are not supported by completed interviews and user studies. They form the basis for the existing process analysis; weaknesses identified in section 3.5 and the requirements for the new system proposed in section 3.6.

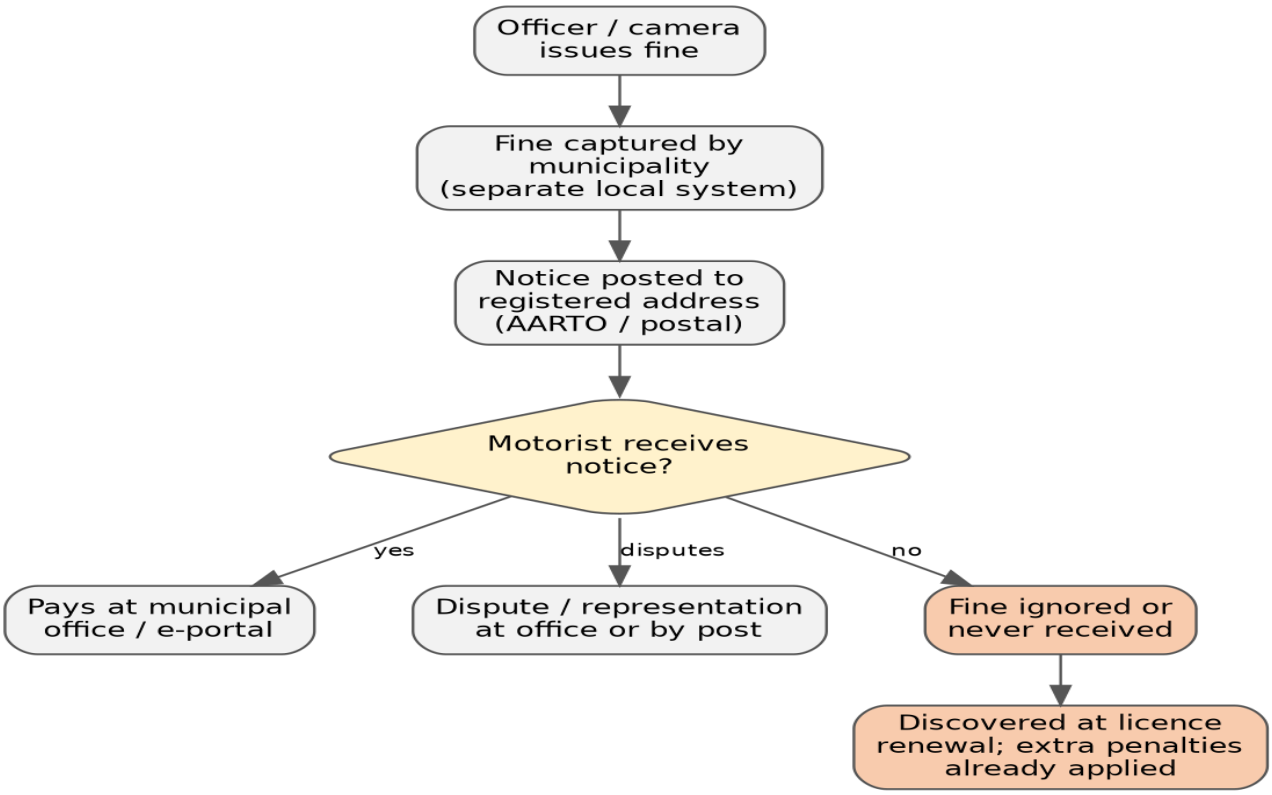
3.3 Analysis of the Existing System.

The existing arrangement is not one system but several loosely related ones in which individual municipalities and metropolitan police departments each run their own fine-management systems which are supported by the national AARTO framework and the eNatis licensing environment. These do not share a common, ID-based record of a motorist's fines.

3.3.1 How the Current System Works?

Step	Activity	Description
1	**Offence and issuing**	A traffic officer or camera records the offence. Details are captured on a notice or into the issuing authority's own system
2	**Capturing**	The fine is stored in that municipality's database. It is linked to a vehicle or address, not to a single national motorist record.
3	**Notification**	A notice is posted to the address registered against the vehicle or motorist. There is no instant electronic alert with photographic evidence.
4	**Payment**	The motorist pays at a municipal office, bank or an e-services portal. Each authority has its own payment channel and reference format.
5	**Dispute/Representation**	A motorist who disagrees must make representation through the prescribed process, often in person or by post, with supporting documents.
6	**Follow-up**	Unpaid fines escalate through further notices and penalties. The motorist frequently learns of the problem only when renewing the driver's license.

Below is the current fine-handling process:



3.3.2 Users and Stakeholders of the Current System

Stakeholder	Role and interest
Motorist	Receives, pays or disputes fines. Needs timely, accurate information.
Traffic officer	Issues fines and records offence details.
Municipal fines / back-office clerk	Captures fines, processes payments and handles representations.
Municipality / traffic authority	Collects revenue and must ensure fines are enforceable and traceable.
Licensing department	Checks for outstanding fines when a licence is renewed.

3.3.3 Information Flow Observed

Information flows one way (authority to motorist) by post and returns only when the motorist acts.

There is no automatic feedback if a notice is undelivered, no automatic reminder, and no shared record between authorities.

Fines also remain invisible to the motorist until a notice arrives or the motorist makes an enquiry.

3.4 Data Analysis (Data Integrity and Constraints)

Data analysis identifies the information that the NatiFine SA prototype must store, the sources of that information and the rules required to maintain its accuracy, consistency and security. The system stores motorist details, traffic-fine records, payment information, dispute submissions and notification histories. Appropriate integrity rules are necessary to prevent duplicate records, invalid data and unauthorised access.

The project is an academic prototype and will use fictitious information rather than actual motorists' personal information or live government records. The data model is designed to support the functions identified in Phase 2, including viewing fines, simulating payments, submitting disputes and generating reminders. Data protection is also considered in line with the Protection of Personal Information Act 4 of 2013 (POPIA), which establishes requirements for the lawful processing and protection of personal information (Africa, 2013).

3.4.1 Data Sources and Data Classification

The data required by NatiFine SA will originate from the following sources:

- **Motorist information:** The motorist enters registration details such as an ID number, name, surname, telephone number, email address and password. Fictitious details will be used during development and demonstrations.
- **Fine information:** An authorised officer or administrator captures the fine identification number, motorist reference, issue date, due date, offence description, location, amount and photograph reference.
- **Payment information:** The simulated payment module records the amount, payment date, payment method and transaction status. No actual banking transactions will take place.
- **Dispute information:** The motorist provides the reason for disputing a fine and may attach supporting evidence. The administrator updates the dispute status.
- **Notification information:** The system generates simulated SMS and WhatsApp-style messages and stores their type, content and date in the notification history.
- **Security and access information:** The system stores account identifiers, password hashes and user roles to support authentication and access control.

The data can be classified as personal, operational, financial, dispute-related and security information. Access must be restricted according to the user's role. For example, motorists should only access their own fine records, while authorised officers or administrators may access the records required to perform their assigned functions.

The prototype will follow the principles of data minimisation and reasonable security safeguards. POPIA requires appropriate technical and organisational measures to protect personal information against loss, damage and unauthorised access or

processing (Africa, 2013).

3.4.2 Data Integrity Rules

The following rules will be applied to maintain data accuracy and consistency.

Entity integrity: Each database table must have a primary key that uniquely identifies every record. The primary keys are UserID, OfficerID, FineID, PaymentID, DisputeID and NotificationID. These keys must not be null or duplicated.

Referential integrity: Foreign keys must refer to existing records. Fine.UserID must reference an existing motorist, and Fine.OfficerID must reference an existing officer. Payment.FineID, Dispute.FineID and Notification.FineID must reference an existing fine. These rules prevent orphan records. Foreign-key enforcement must be enabled in SQLite.

Domain integrity: Each field must contain data of the correct type and format. For example, fine amounts must be positive numbers, email addresses must follow an acceptable format, dates must be valid and status fields must contain approved values.

Business-rule integrity: The system must prevent a motorist from paying the same fine more than once. In the prototype, each fine may have no more than one full payment record and no more than one dispute record. A fine with an open dispute must not receive an additional prototype penalty while the dispute is being reviewed. Once a payment is successful, the fine status must be updated to Paid.

Security and confidentiality: Passwords must be stored as secure password hashes rather than plain text. Role-based access controls must prevent motorists from accessing another motorist's information or performing officer-only functions. These safeguards support the protection of personal information required by POPIA.

Auditability: Important actions, including fine creation, payment attempts, dispute decisions and changes to licence-status flags, should be recorded with dates and times. Fine records should be marked as cancelled or otherwise updated where appropriate instead of being physically deleted, so that the prototype can preserve a history of relevant actions.

3.4.3 Data Constraints

The database will use the following constraints to reduce invalid or inconsistent records:

- **User.IDNumber:** Required and unique, with exactly 13 digits for the South African ID-number format used by the prototype. Validation must check the format and, where implemented, the applicable checksum. Fictitious test values will be used.
- **User.Phone:** Required and validated against an accepted South African mobile-number format.

- **User.Email:** Required, unique and checked for an acceptable email format.
- **User.PasswordHash:** Required and stored as a password hash using an appropriate password-hashing method.
- **User.LicenceStatus:** Restricted to the prototype values Valid and Flagged.
- **Fine.Amount:** Required and greater than zero.
- **Fine.PenaltyAmount:** Must not be negative and will default to zero.
- **Fine.IssueDate and Fine.DueDate:** Both required, with the due date later than the issue date and the issue date not in the future.
- **Fine.Status:** Restricted to Unpaid, Paid, Disputed, Overdue or Cancelled.
- **Payment.FineID:** A foreign key and unique value in the prototype, ensuring no more than one payment record per fine.
- **Dispute.FineID:** A foreign key and unique value in the prototype, ensuring no more than one dispute record per fine.
- **Notification.FineID:** A foreign key referencing an existing fine; multiple notifications may be associated with one fine.

These constraints must be implemented in the database where possible, supported by application-level validation and verified through testing. The detailed field definitions are provided in the data dictionary in Section 3.8.

3.4.4 Prototype Business Rules and Assumed Thresholds

The reminder, penalty and licence-status rules are configurable assumptions introduced for the NatiFine SA academic prototype. They are not presented as official South African traffic-fine deadlines, statutory penalty amounts or legal licence-disqualification rules.

For demonstration purposes, the proposed system will apply the following rules:

1. **First reminder:** A simulated reminder is generated when a fine remains unpaid for 14 days after its issue date.
2. **Additional penalty:** A configurable prototype penalty is applied once when a fine remains unpaid for 30 days after its issue date. The penalty amount and calculation method must be defined in the system configuration.
3. **Licence-status flag:** The motorist's prototype licence-status field is changed to Flagged when an eligible fine remains unpaid for 60 days after its issue date. This is only an internal demonstration flag and does not represent an actual change to a driver's legal licence status.
4. **Disputed fines:** An additional prototype penalty is not applied while a dispute remains open. When the dispute is resolved, the fine's status and any subsequent processing must follow the configured prototype rules.

5. **Successful payment:** Once a simulated payment succeeds, the fine is marked Paid, the payment record is retained, and no further unpaid-fine reminders or additional penalties are generated for that fine.
6. **Status restoration:** The prototype licence-status flag is cleared when the motorist has no remaining eligible overdue fines under the configured rules.

The thresholds will be stored as configurable values so that they can be changed without redesigning the database. Actual traffic-fine procedures and applicable deadlines depend on the relevant legal framework and circumstances. The Department of Transport has also published implementation updates concerning the phased implementation of AARTO, demonstrating why the prototype's simplified rules should not be treated as a substitute for current official procedures (Government, 2025).

These business rules provide a consistent basis for demonstrating reminders, simulated penalties and status updates during development and testing. All such actions will occur within the prototype and will not affect real fines, bank accounts or driving licences.

3.4.5 Weaknesses of the Current System

The following weaknesses were identified from the observation, participation and interview findings and the Phase 2 investigation. Each is linked to the requirement in section 3.6 that addresses it.

No.	Weakness	Effect	Addressed by
1	Fragmented systems	Each municipality keeps its own records, so there is no complete view of a motorist's fines.	FR2, FR3
2	Fines linked to address, not ID	Fines are posted to outdated or wrong addresses and never reach the motorist.	FR1, FR3
3	Late discovery of fines	Motorists often learn of fines only at licence renewal, after penalties have accumulated.	FR4, FR8
4	Slow, postal notification	There is delay between the offence and the motorist knowing; no photographic evidence is attached.	FR4
5	Manual or in-person payment	Queues, travel and office hours inconvenience motorists and burden staff.	FR5

No.	Weakness	Effect	Addressed by
6	Difficult dispute process	Representations are paper-based and hard to track, leaving motorists without status updates.	FR6
7	No automatic reminders or escalation	Unpaid fines depend on manual follow-up, so revenue is lost and penalties are inconsistent.	FR7, FR8, FR9
8	Poor visibility for authorities	Staff cannot easily see which notices were received, paid or contested.	FR10
9	Data quality and duplication risk	Inconsistent formats between systems lead to duplicate or mismatched records.	Section 3.4 constraints

3.4 Analysis of the Proposed System

NatiFine SA replaces the fragmented, address-based process with one centralized web application in which all fines are held against the motorist's ID number. This section lists what the system must do. Requirement identifiers FR1 to FR12 are used so that they can be traced through design, development and testing stages.

3.6.1 Actors

Actor	Description
Motorist	Registered citizen who views, pays and disputes fines and receives notifications.
Traffic Officer / Admin	Authorised user who captures fines against a motorist's ID number and reviews disputes.
System (scheduler)	Automated process that sends reminders, applies penalties and updates licence status.
Simulated payment / messaging services	Internal simulations that stand in for banks and SMS / WhatsApp providers.

3.4.2 Functional Requirement

ID	Requirement	Description	Priority
FR1	Registration and authentication	A motorist can register with ID number, name, surname, phone, email and password, and log in. Passwords are hashed. Officers/admins log in separately with role-based access.	Must have
FR2	View fines	A logged-in motorist sees all fines linked to their ID number showing date, location, amount, penalty, status and photo, with filtering by status.	Must have
FR3	ID-based centralised record	All fines are stored in one database and found by ID number, regardless of the issuing officer or location.	Must have
FR4	Fine notification	When a fine is added, the system generates a simulated SMS and a WhatsApp-style message with the fine photo and logs it in the notification history.	Must have
FR5	Online payment	A motorist selects an unpaid fine, confirms the amount (including any penalty) and completes a simulated payment. The system issues a receipt and marks the fine Paid.	Must have
FR6	Dispute a fine	A motorist submits a dispute form for a fine with a reason and optional evidence, and can track its status (Submitted, Under Review, Upheld, Rejected).	Must have
FR7	Automatic reminders	The scheduler sends reminders for fines unpaid after the configured period and records each reminder.	Must have
FR8	Licence status flag	If a fine remains unpaid beyond the threshold, the motorist's LicenceStatus is set to Flagged, and cleared when the fines are settled.	Must have
FR9	Additional penalties	When the threshold passes, a penalty amount is recorded against the fine once and shown to the motorist.	Must have

ID	Requirement	Description	Priority
FR10	Officer / admin interface	An authorised user can capture new fines by ID number, view fines and payments, and update dispute decisions.	Must have
FR11	Receipts and payment history	A motorist can view and print receipts for paid fines.	Should have
FR12	Profile management	A motorist can update phone and email and change their password.	Should have

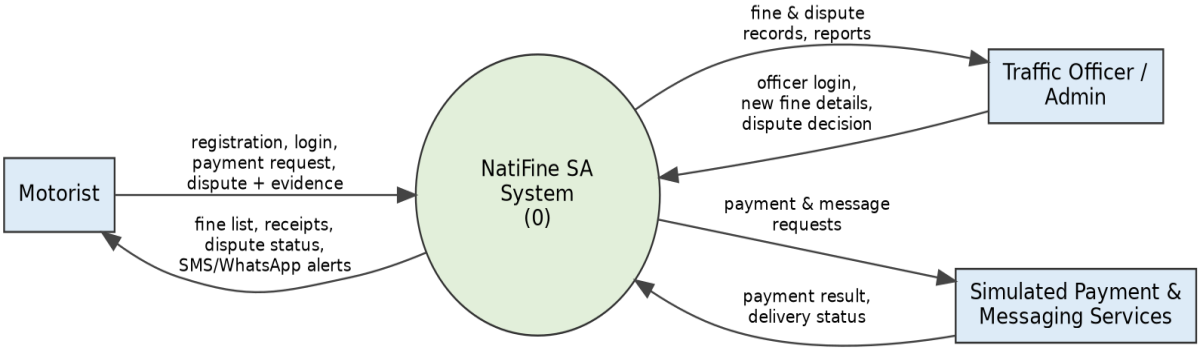
3.4.3 Use Case Overview

Use case	Actor	Brief flow
Register / Log in	Motorist, Officer	Enter credentials → system validates → session started.
Capture fine	Officer / Admin	Enter ID number and offence details → system validates → fine stored → notification triggered.
View fines	Motorist	Log in → open “My Fines” → system returns fines for that ID.
Pay fine	Motorist	Select fine → confirm amount → simulated payment → receipt → status Paid.
Dispute fine	Motorist	Select fine → submit reason and evidence → status Disputed → admin decision.
Review dispute	Officer / Admin	Open dispute → view evidence → record decision → motorist notified.
Send reminders / apply penalties	System	Scheduler checks fine ages → sends reminder, adds penalty, flags licence as thresholds are reached.

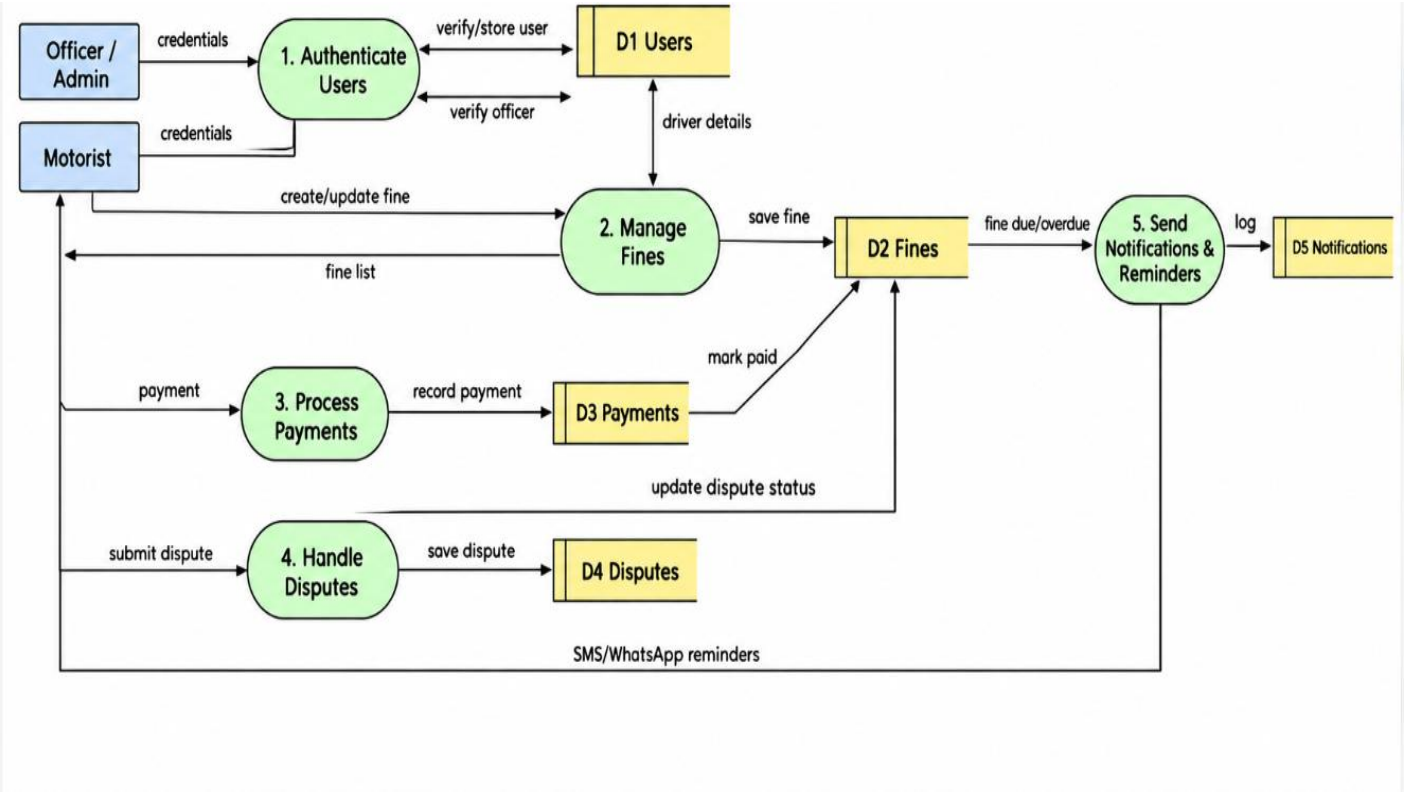
3.4.4 Data Flow Analysis

The context diagram shows the system as a single process and its interaction with external entities. Level 1 decomposes it into five processes and six data stores that correspond to the entities in section 3.8.

This is the context diagram:



Below is the Data Flow Diagram:



3.4.5 Assumptions and Constraints Carried Forward

The assumptions and constraints from Phase 2 continue to apply: users have basic digital skills and access to a phone or computer; all data is fictitious; the project runs on a 10-week schedule with three developers; and no paid infrastructure or third-party notification services are used. Live government, banking and SMS / WhatsApp integration is out of scope and is simulated.

3.5 Non-Functional Requirements

Non-functional requirements describe the quality attributes of the system. Those listed in Phase 2 are retained and are here made measurable so that they can be verified during the Testing sprint.

ID	Attribute	Requirement	How it will be verified
NFR1	Usability	A first-time user can view and pay a fine within a few clicks (no more than five steps from login). Layouts are consistent, responsive, and use clear labels and error messages.	Walkthrough with test users; count of steps
NFR2	Performance	A user's fine list loads in under 2 seconds on the prototype database; payment and dispute submission respond within 3 seconds.	Timed tests with sample data
NFR3	Security	Passwords are salted and hashed. Only the authenticated owner can view or pay their fines. Officer functions require the officer role. Inputs are validated and queries are parameterized to prevent SQL injection, and output is escaped to prevent XSS. Sessions expire after inactivity.	Test cases on access control and invalid input
NFR4	Reliability	The prototype runs through the full demo scenario (register, view, pay or dispute) without crashing. Failed payments or errors leave data in a consistent state.	End-to-end test; error-case tests
NFR5	Availability	Available during development, testing and demonstration sessions rather than 24/7.	Demo-day readiness check

ID	Attribute	Requirement	How it will be verified
NFR6	Maintainability	Code is organized into modules (authentication, fines, payments, notifications) with comments and naming standards so any team member can maintain it.	Code review
NFR7	Data integrity	All constraints in section 3.4 are enforced at database level, not only in the user interface.	Attempt to insert invalid data
NFR8	Compatibility / portability	Works in current versions of Chrome, Edge and Firefox on desktop and mobile screens, with no installation beyond a browser.	Cross-browser check
NFR9	Privacy	Only the minimum personal data needed is stored and is displayed only to its owner and authorized officers, in the spirit of POPIA. Prototypes use fictitious data only.	Review of stored fields
NFR10	Auditability	Key events (fine creation, payment, dispute decision, license flag change) are time-stamped so history can be traced.	Inspect records after test run

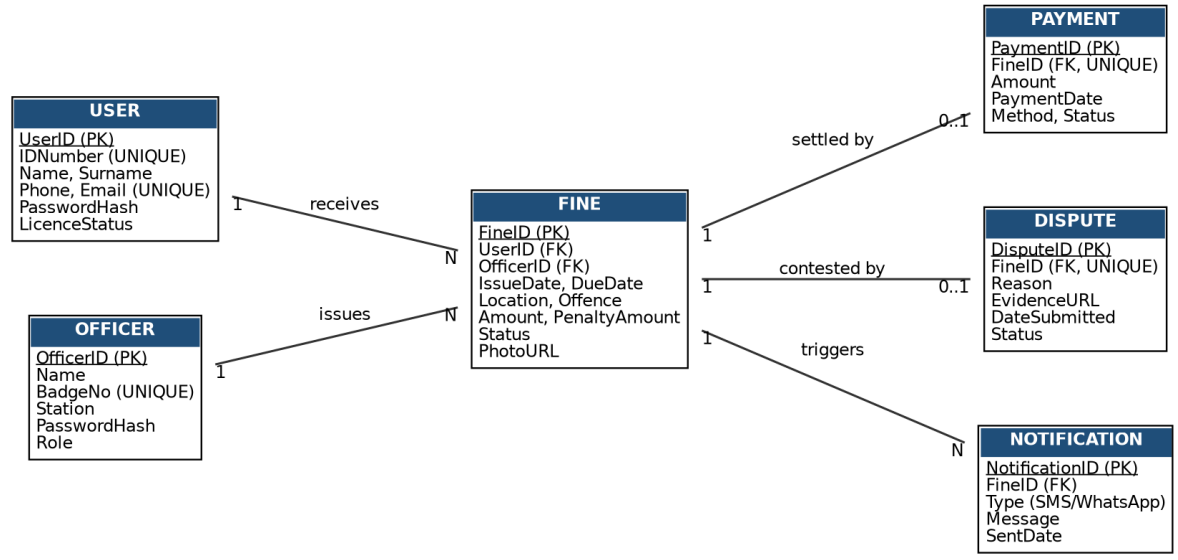
3.8 Data Modelling

Phase 2 listed the principal entities. This section expands them into a full entity relationship model, a data dictionary and a normalization check. It also corrects the Officer/Admin relationship, which Phase 2 described incorrectly: the relationship is that one officer issues many fines, and each fine is issued by exactly one officer.

3.8.1 Entities and Relationships

Relationship	Cardinality	Explanation
User receives Fine	1 : N	One motorist can have many fines. Each fine belongs to exactly one motorist.
Officer issues Fine	1 : N	One officer can issue many fines. Each fine records exactly one issuing officer.
Fine settled by Payment	1 : 0.1	A fine has no payment or one full payment in the prototype.
Fine contested by Dispute	1 : 0.1	A fine has no dispute or one dispute. A user may submit multiple disputes through their different fines.
Fine triggers Notification	1 : N	A fine can have many notifications (initial alert, reminders, payment confirmation).

Entity Relationship Diagram for NatiFine SA.



3.8.2 Data Dictionary

Types are give for SQLite, PK= Primary Key, FK = Foreign Key

USER (Motorist)

Attribute	Type	Key	Notes
UserID	INTEGER	PK	Auto-increment
IDNumber	TEXT(13)	UNIQUE	South African ID number
Name	TEXT		Not null
Surname	TEXT		Not null
Phone	TEXT		Not null; used for notifications
Email	TEXT	UNIQUE	Valid email
PasswordHash	TEXT		Salted hash
LicenceStatus	TEXT		Valid / Flagged, default Valid

OFFICER

Attribute	Type	Key	Notes
OfficerID	INTEGER	PK	Auto-increment
Name	TEXT		Not null
BadgeNo	TEXT	UNIQUE	Badge / employee number
Station	TEXT		Issuing station or department
PasswordHash	TEXT		Salted hash (added to support officer login, FR1 and FR10)
Role	TEXT		Officer / Admin

FINE

Attribute	Type	Key	Notes
FineID	INTEGER	PK	Auto-increment
UserID	INTEGER	FK	References User
OfficerID	INTEGER	FK	References Officer
IssueDate	DATE		Not null, not in the future
DueDate	DATE		Not null, after IssueDate (added to drive reminders)
Location	TEXT		Where the offence occurred

Attribute	Type	Key	Notes
Offence	TEXT		Offence description (added for clarity)
Amount	DECIMAL(10,2)		Greater than 0
PenaltyAmount	DECIMAL(10,2)		Default 0
Status	TEXT		Unpaid / Paid / Disputed / Overdue / Cancelled
PhotoURL	TEXT		Path to the fine photograph

PAYMENT

Attribute	Type	Key	Notes
PaymentID	INTEGER	PK	Auto-increment
FineID	INTEGER	FK, UNIQUE	References Fine; one payment per fine
Amount	DECIMAL(10,2)		Amount paid
PaymentDate	DATETIME		Time of simulated payment
Method	TEXT		Simulated: Card / EFT
Status	TEXT		Successful / Failed

DISPUTE

Attribute	Type	Key	Notes
DisputeID	INTEGER	PK	Auto-increment
FineID	INTEGER	FK, UNIQUE	References Fine; one dispute per fine
Reason	TEXT		Not null
EvidenceURL	TEXT		Optional uploaded evidence
DateSubmitted	DATETIME		Not null
Status	TEXT		Submitted / Under Review / Upheld / Rejected

NOTIFICATION

Attribute	Type	Key	Notes
NotificationID	INTEGER	PK	Auto-increment
FineID	INTEGER	FK	References Fine
Type	TEXT		SMS / WhatsApp
Message	TEXT		Message content
SentDate	DATETIME		Time sent (simulated)

3.8.3 Normalization Check

First Normal Form (1NF): every attribute holds a single atomic value and every table has a primary key; there are no repeating groups (multiple notifications and payments are separate rows in their own tables).

Second Normal Form (2NF): all tables use a single-column primary key, so no non-key attribute depends on only part of a key.

Third Normal Form (3NF): non-key attributes depend only on the key. Motorist details live only in User, officer details only in Officer, and Fine stores references (UserID, OfficerID) instead of copying names. The model is therefore in 3NF.

Design note. PenaltyAmount is stored on Fine as required in Phase 2. It is derived from the Amount and the penalty rule, so it is deliberately recorded as a historical value (the penalty applied at that time) rather than recalculated, which protects the record if penalty rules change later.

3.8.4 Changes from the Phase 2 Data Model

Change	Reason
Added PasswordHash and Role to Officer	Officers and admins must authenticate (FR1, FR10, NFR3).
Added DueDate and Offence to Fine	Needed for reminder timing and to describe the fine (FR2, FR7).
Corrected the Officer–Fine relationship to 1 : N	The Phase 2 wording referred to disputes, but the officer's relationship is to the fines issued.
Added UNIQUE constraints on Payment.FineID and Dispute.FineID	Enforces the 0.1 relationships at database level.

Conclusion

The Analysis Phase has shown how fines are handled today, why that process fails motorists and authorities, and what NatiFine SA must do to fix it. Twelve functional requirements, ten non-functional requirements, a set of integrity constraints and a normalized six-entity data model now define the prototype precisely. These outputs feed directly into the Design phase, where screens, database scripts and module structure will be produced, and into the Testing sprint, where each requirement will be checked.

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